

THE PRIMITIVE BRAIN

IN TIMES OF STRESS OR CRISIS, DEEP-SEATED INSTINCTS WELL UP FROM WITHIN US AND TAKE OVER OUR AWARENESS. SUCH EVENTS INVOLVE THE “PRIMITIVE BRAIN”, WHICH IS BASED MAINLY IN A SERIES OF PARTS KNOWN AS THE LIMBIC SYSTEM.

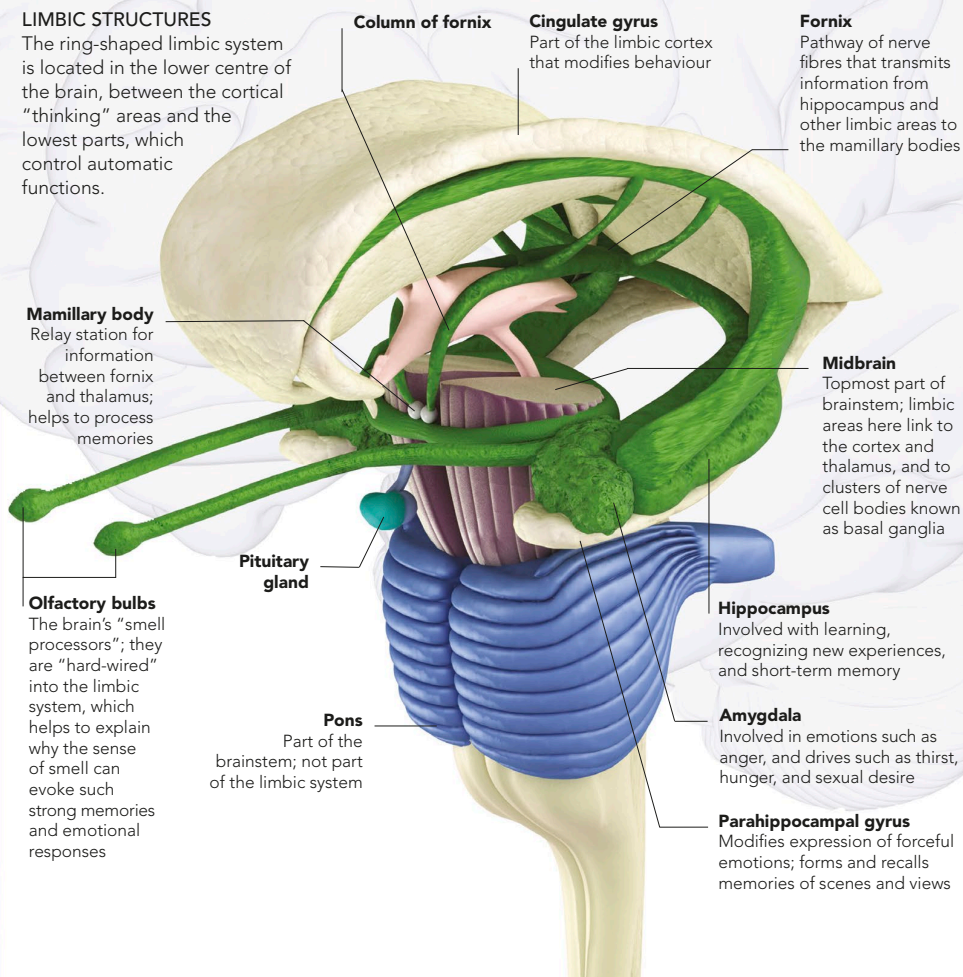
THE LIMBIC SYSTEM

The limbic system influences subconscious, instinctive behaviour, similar to animal responses that relate to reproduction and survival. In humans, many of these innate, “primitive” behaviours are modified by conscious, thoughtful considerations based

in upper regions of the brain. However, when primal urges prevail, the limbic system and its associated structures take over. At other times they play lesser, but still complex and important, roles in the expression of instincts, drives, and emotions.

LIMBIC STRUCTURES

The ring-shaped limbic system is located in the lower centre of the brain, between the cortical “thinking” areas and the lowest parts, which control automatic functions.



Mamillary body

Relay station for information between fornix and thalamus; helps to process memories

Olfactory bulbs

The brain's “smell processors”; they are “hard-wired” into the limbic system, which helps to explain why the sense of smell can evoke such strong memories and emotional responses

Column of fornix

Cingulate gyrus

Part of the limbic cortex that modifies behaviour

Fornix

Pathway of nerve fibres that transmits information from hippocampus and other limbic areas to the mamillary bodies

Midbrain

Topmost part of brainstem; limbic areas here link to the cortex and thalamus, and to clusters of nerve cell bodies known as basal ganglia

Hippocampus

Involved with learning, recognizing new experiences, and short-term memory

Amygdala

Involved in emotions such as anger, and drives such as thirst, hunger, and sexual desire

Parahippocampal gyrus

Modifies expression of forceful emotions; forms and recalls memories of scenes and views

Pons

Part of the brainstem; not part of the limbic system

Pituitary gland